

# Capital Structure Decision

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# Sources of Long-Term Finance

## Debt

- Returns are fixed
- Debt interest is tax deductible
- Interest is an obligation that cannot be avoided

## Equity

- Returns are dividends
- Volatile and uncertain
- High rates of return to compensate for high risk

# Equity Financing Options



Rights issue to existing  
shareholders

Public issue of shares

Private placing

- Bond Issue
- Debenture Issue
- Convertible Bond Issue
- Syndicated Loan

# Debt Financing Options

# Weighted Average Cost of Capital (WACC)

## Formula

$$WACC = \left( \frac{V_e}{V_e + V_d} \right) k_e + \left( \frac{V_d}{V_e + V_d} \right) k_d (1 - T)$$

$V_e$  = market values of equity

$V_d$  = market values of debt

$k_e$  = cost of equity

$k_d(1 - T)$  = cost of debt

$T$  = corporate tax rate

# Cost of Equity

**CAPM**

Capital Asset  
Pricing Model

**DGM**

Dividend  
Growth  
Model

**EGM**

Earnings  
Growth  
Model

# Capital Asset Pricing Model (CAPM)

$$\text{Cost of Equity} = R_f + \beta \times (R_m - R_f)$$

$R_f$  = Risk free rate

$R_m$  = Market return

$\beta$  = Systematic risk / Variation

- Level of systematic risk faced by an investor
- Beta > 1 indicates above average risk
- Beta < 1 means relatively low risk
- Beta = Covariance (Rp,Rm) / Variance (Rm)
- Proxy Beta
- Beta values for companies reflect both:
  - business risk (resulting from operations)
  - finance risk (resulting from their level of gearing)

# Beta Factor

# Leverage



## Operating Leverage

Firms ability to use fixed operating costs to magnify effects of changes in Sales on its EBIT

Degree of Operating Leverage:

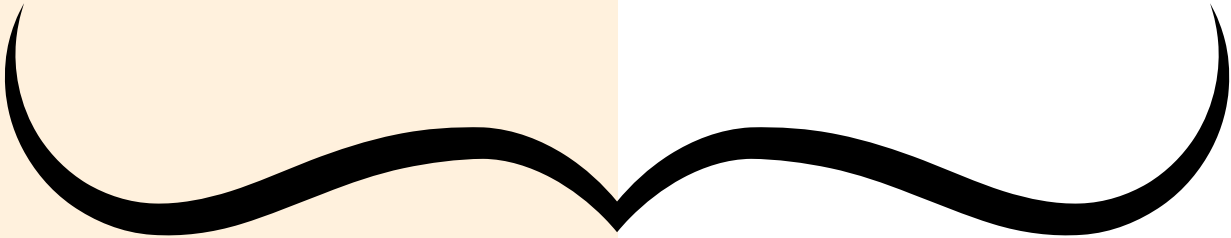
$$\text{DOL} = \frac{\% \text{ Change in EBIT}}{\% \text{ Change in Sales}}$$

Firm's ability to use fixed financial changes (interest) to magnify the effect of changes in EBIT on the firm's EPS.

Degree of Financial Leverage:

$$\text{DFL} = \frac{\% \text{ Change in EPS}}{\% \text{ Change in EBIT}}$$

## Financial Leverage



## Combined Leverage

Degree of Combined Leverage:

$$\text{DCL} = \frac{\% \text{ Change in EPS}}{\% \text{ Change in Sales}}$$

# Question A

Sunrise Manufacturing Ltd. produces and sells 50,000 units of its product annually. The selling price per unit is Rs. 150. The variable cost per unit is Rs. 90, and the company incurs total fixed costs of Rs. 20,00,000.

The company has an outstanding debt of Rs. 10,00,000, on which it pays an annual interest rate of 10%. Sunrise Manufacturing Ltd. has 10,00,000 equity shares outstanding. The corporate tax rate applicable to the company is 30%.

Based on the above information, calculate the following for Sunrise Manufacturing Ltd.:

1. Degree of Operating Leverage (DOL)
2. Degree of Financial Leverage (DFL)
3. Degree of Combined Leverage (DCL)

|                                                            | Units Sold = 50,000 | Units Sold = 60,000 |
|------------------------------------------------------------|---------------------|---------------------|
| <b>Sales</b> = Units Sold x Sales price per unit           | 75,00,000           | 90,00,000           |
| <b>Variable Cost</b> = Units Sold x Variable cost per unit | 45,00,000           | 54,00,000           |
| <b>Contribution</b> = Sales - Variable Cost                | 30,00,000           | 36,00,000           |
| <b>Fixed Cost</b>                                          | 20,00,000           | 20,00,000           |
| <b>EBIT</b> = Contribution - Fixed Cost                    | 10,00,000           | 16,00,000           |
| <b>Interest</b> = Cost of debt x Total debt                | 1,00,000            | 1,00,000            |
| <b>EBT</b> = EBIT - Interest                               | 9,00,000            | 15,00,000           |
| <b>Taxes</b> = EBT x Tax rate                              | 2,70,000            | 4,50,000            |
| <b>PAT</b> = EBT - Taxes                                   | 6,30,000            | 10,50,000           |
| <b>EPS</b> = PAT / Number of shares                        | 0.63                | 1.05                |

$$\% \text{ change in EBIT} = (16,00,000 - 10,00,000) / 10,00,000 = 0.6 = 60\%$$

$$\% \text{ change in Sales} = (90,00,000 - 75,00,000) / 75,00,000 = 0.2 = 20\%$$

$$\% \text{ change in EPS} = (1.05 - 0.63) / 0.63 = 0.667 = 66.7\%$$

$$\begin{aligned} \text{DOL} &= \% \text{ change in EBIT} / \% \text{ change in Sales} \\ \text{DOL} &= 0.6 / 0.2 = 3.00 \end{aligned}$$

$$\begin{aligned} \text{DFL} &= \% \text{ change in EPS} / \% \text{ change in EBIT} \\ \text{DFL} &= 0.667 / 0.6 = 1.11 \end{aligned}$$

$$\begin{aligned} \text{DCL} &= \% \text{ change in EPS} / \% \text{ change in Sales} \\ \text{DCL} &= 0.667 / 0.2 = 3.33 \end{aligned}$$

$$\begin{aligned} \text{DOL} &= \text{Contribution} / \text{EBIT} \\ \text{DOL} &= 30,00,000 / 10,00,000 = 3.00 \end{aligned}$$

$$\begin{aligned} \text{DFL} &= \text{EBIT} / \text{EBT} \\ \text{DFL} &= 10,00,000 / 9,00,000 = 1.11 \end{aligned}$$

$$\begin{aligned} \text{DCL} &= \text{Contribution} / \text{EBT} \\ \text{DCL} &= 30,00,000 / 9,00,000 = 3.33 \end{aligned}$$

## Operating Leverage

Degree of Operating Leverage:

$$DOL = \frac{\text{Contribution}}{EBIT}$$

## Financial Leverage

Degree of Financial Leverage:

$$DFL = \frac{EBIT}{EBT}$$

## Combined Leverage

Degree of Combined Leverage:

$$DCL = \frac{\text{Contribution}}{EBT}$$

# Question B

Blue Horizon Ltd. is involved in the manufacturing and sale of a product, with an annual output of 80,000 units. Each unit is sold for Rs. 200, while the variable cost associated with each unit is Rs. 120. The company incurs fixed costs totaling Rs. 30,00,000.

To finance its operations, Blue Horizon Ltd. has taken on debt amounting to Rs. 15,00,000, which carries an interest rate of 8% per annum. The company is also financed by equity, with a total of 2,00,000 shares outstanding. The applicable corporate tax rate for Blue Horizon Ltd. is 25%.

Given this information, you are required to:

1. Calculate the Degree of Operating Leverage (DOL).
2. Determine the Degree of Financial Leverage (DFL).
3. Compute the Degree of Combined Leverage (DCL).

|                                                            | Amount      |
|------------------------------------------------------------|-------------|
| <b>Sales</b> = Units Sold x Sales price per unit           | 1,60,00,000 |
| <b>Variable Cost</b> = Units Sold x Variable cost per unit | 96,00,000   |
| <b>Contribution</b> = Sales - Variable Cost                | 64,00,000   |
| <b>Fixed Cost</b>                                          | 30,00,000   |
| <b>EBIT</b> = Contribution - Fixed Cost                    | 34,00,000   |
| <b>Interest</b> = Cost of debt x Total debt                | 1,20,000    |
| <b>EBT</b> = EBIT - Interest                               | 32,80,000   |

$$\text{DOL} = \text{Contribution} / \text{EBIT}$$

$$\text{DOL} = 64,00,000 / 34,00,000$$

$$\text{DOL} = 1.88$$

$$\text{DFL} = \text{EBIT} / \text{EBT}$$

$$\text{DFL} = 34,00,000 / 32,80,000$$

$$\text{DFL} = 1.04$$

$$\text{DFL} = \text{Contribution} / \text{EBT}$$

$$\text{DFL} = 64,00,000 / 32,80,000$$

$$\text{DFL} = 1.95$$

# EBIT-EPS Analysis

**Technique to identify the capital structure which gives maximum benefit to owners (EPS)**

# Question C

Sunshine Ltd. is considering two financing plans to fund a new project. The required capital is Rs. 20,00,000. The details of the two plans are as follows:

- Plan X: Issue 2,00,000 equity shares at Rs. 10 each.
- Plan Y: Issue 1,00,000 equity shares at Rs. 10 each and raise Rs. 10,00,000 in debt at an interest rate of 8% per annum.

The expected EBIT from the new project is Rs. 5,00,000. The corporate tax rate is 30%.

For each financing plan, calculate the EPS and advise which plan Sunshine Ltd. should choose based on EBIT-EPS analysis.

|                  | Plan X   | Plan Y   |
|------------------|----------|----------|
| Number of shares | 2,00,000 | 1,00,000 |
| Share price      | 10       | 10       |
| Debt             | 0        | 1,00,000 |
| Cost of debt     | 0        | 8%       |
| EBIT             | 5,00,000 | 5,00,000 |
| Tax rate         | 30%      | 30%      |
|                  |          |          |
| EBIT             | 5,00,000 | 5,00,000 |
| Interest         | 0        | 80,000   |
| EBT              | 5,00,000 | 4,20,000 |
| Taxes            | 1,50,000 | 1,26,000 |
| PAT              | 3,50,000 | 2,94,000 |
| EPS              | 1.75     | 2.94     |

# Question D

Suppose a firm has a capital structure exclusively comprising of ordinary shares amounting to Rs. 10,00,000 of Rs. 100 each. The firm now wishes to raise additional Rs. 10,00,000 for expansion. The firm has three alternative financial plans:

1. It can raise the entire capital by issuing equity shares of Rs 100 each.
2. It can raise 50% by issuing Rs. 100 equity shares and 50% by issuing 5% debentures.
3. It can raise the entire amount by using 6% debentures.

Assume the EBIT is Rs. 1,20,000 and tax rate is 30%. Which financing plan should the firm select?

|              | Plan 1   | Plan 2   | Plan 3   |
|--------------|----------|----------|----------|
| Equity %     | 100%     | 50%      | 0%       |
| Debt %       | 0%       | 50%      | 100%     |
| Cost of debt |          | 5%       | 6%       |
| EBIT         | 1,20,000 | 1,20,000 | 1,20,000 |
| Tax rate     | 30%      | 30%      | 30%      |

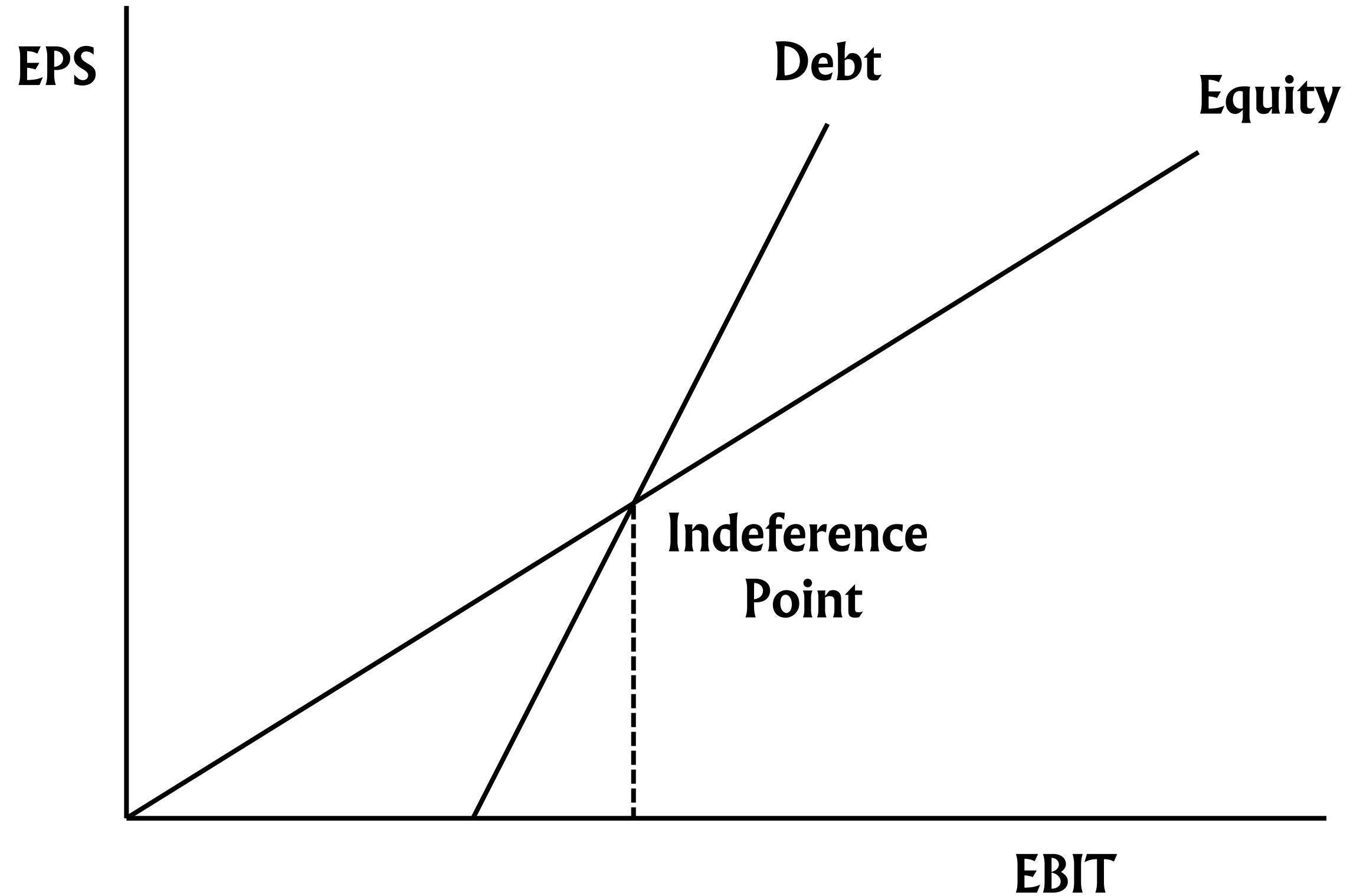
|                                                             |           |           |           |
|-------------------------------------------------------------|-----------|-----------|-----------|
| Total equity = Current Equity + Equity % x Additional funds | 20,00,000 | 15,00,000 | 10,00,000 |
| Share price                                                 | 100       | 100       | 100       |
| Total Debt = Debt % x Additional funds                      | 0         | 5,00,000  | 10,00,000 |

|                                      |          |          |          |
|--------------------------------------|----------|----------|----------|
| EBIT                                 | 1,20,000 | 1,20,000 | 1,20,000 |
| Interest = Cost of debt x Total Debt | 0        | 25,000   | 60,000   |
| Taxes = (EBiT - Interest) x Tax rate | 36,000   | 28,500   | 18,000   |
| PAT                                  | 84,000   | 66,500   | 42,000   |

|                                               |        |        |        |
|-----------------------------------------------|--------|--------|--------|
| Number of shares = Total Equity / Share price | 20,000 | 15,000 | 10,000 |
| EPS = PAT / Number of shares                  | 4.20   | 4.43   | 4.20   |

# Point of Indifference

Level of EBIT where different financing options (or capital structures) result in the same EPS



# Point of Indifference

Level of EBIT where two different financing options result in the same EPS

$$\frac{(\text{EBIT} - I_1) \times (1 - T)}{N_1} = \frac{(\text{EBIT} - I_2) \times (1 - T)}{N_2}$$

where

- EBIT = Earnings Before Interest and Taxes
- $I_1$  = Interest expense for financing plan 1
- $I_2$  = Interest expense for financing plan 2
- $T$  = Tax rate
- $N_1$  = Number of outstanding shares for financing plan 1
- $N_2$  = Number of outstanding shares for financing plan 2

# Question E

Alpha Ltd. requires funds amounting to \$80 million for its new project. To raise funds, the company has following two alternative:

- (a) To issue Equity Shares (at par) amounting to \$60 million and borrow the balance amount at the interest of 12% p.a., or
- (b) To issue Equity Shares (at par) and 12% Debentures in equal proportion.

The income tax rate is 30%.

Find out the point of indifference between the available two modes of financing and state which options will be beneficial in different situations.

# Optimal Capital Structure

Maximises the shareholder's wealth while minimising the cost of capital

# Capital Structure Theories

## Relevance Theories

Believe and prove that **Capital Structure** is relevant in achieving the goal of value maximisation.

**Net Income Approach**

**Traditional Approach  
(WACC Approach)**

## Irrelevance Theories

State that Capital structure does not influence value of the firm and value of equity.

**Net Operating Income approach**

**MM Approach  
(without tax)**

# Net Income Approach

Maximum possible debt in the Capital Structure results in Minimum Overall Cost of Capital

Assumptions:

- Cost of Debt ( $K_d$ ) and Cost of Equity ( $K_e$ ) are constant
- $K_d < K_e$

## Step 1

Calculate Operating Profits (EBIT)

## Step 2

Calculate Net Income

## Step 3

Calculate Value of Equity

Value of Equity = Net Income /  $K_e$

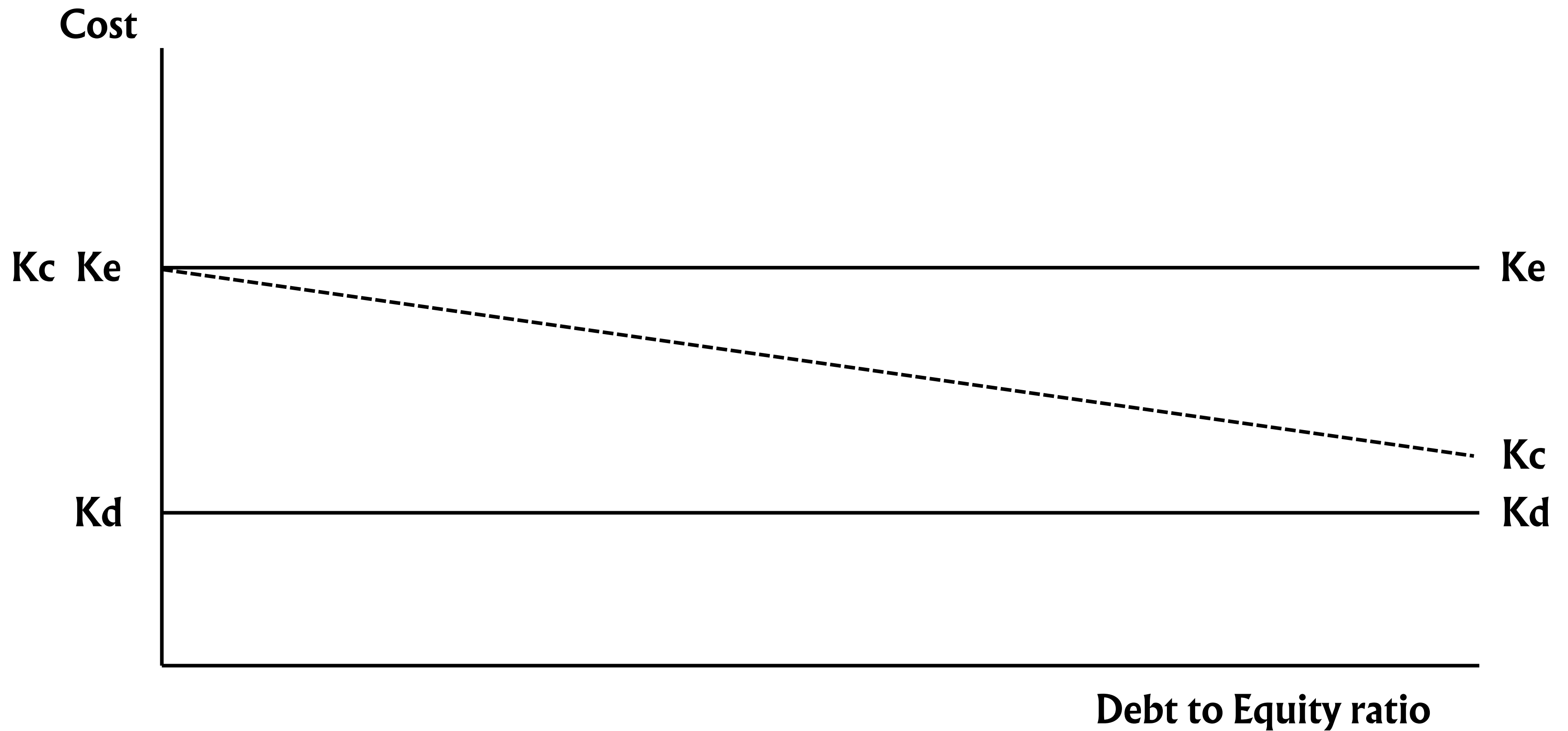
## Step 4

Calculate Value of the Firm

Value of the Firm = Value of Equity + Debt

OR

Value of the Firm = EBIT / WACC



# Net Income Approach

# Question 1

**X Ltd. is expecting an annual EBIT of Rs.1 lakh. The company has Rs.4 lakhs in 10% debentures. The cost of equity capital is 12.5%. You are required to calculate the total value of the firm according to the Net Income Approach.**

**Solution:**

**Net Income = EBIT - Interest**

**Net Income = 1,00,000 - (0.1 x 4,00,000) = 1,00,000 - 40,000 = 60,000**

**Value of Equity = Net Income / Cost of Equity (ke)**

**Value of Equity = 60,000 / 0.125 = 4,80,000**

**Value of the Firm = Value of Equity + Value of Debt**

**Value of the Firm = 4,80,000 + 4,00,000**

**Value of the Firm = 8,80,000**

# Question 2

ABC Ltd earned a profit of Rs.20,00,000 before interest and tax. The Co's Capital structure is as follows

- a) 400,000 equity shares of Rs.10 each with a capitalization rate of 16%
- b) 25,000, 14% debentures of Rs.150 each.

You are required to compute the following using NI approach

- (i) The overall cost of capital
- (ii) Market Value of the firm.

Solution:

$$\text{Value of Debt} = 25,000 \times 150 = 37,50,000$$

$$\text{Net Income} = \text{EBIT} - \text{Interest}$$

$$\text{Net Income} = 20,00,000 - (0.14 \times 37,50,000) = 20,00,000 - 5,25,000 = 14,75,000$$

$$\text{Value of Equity} = \text{Net Income} / \text{Cost of Equity (ke)}$$

$$\text{Value of Equity} = 14,75,000 / 0.16 = 92,18,750$$

$$\text{Market Value of the Firm} = \text{Value of Equity} + \text{Value of Debt}$$

$$\text{Market Value of the Firm} = 92,18,750 + 37,50,000$$

$$\text{Market Value of the Firm} = 1,29,68,750$$

$$\text{Market Value of the Firm} = \text{EBIT} / \text{WACC}$$

$$\text{WACC} = \text{EBIT} / \text{Market value of the Firm}$$

$$\text{WACC} = 20,00,000 / 1,29,68,750$$

$$\text{WACC} = 0.1542 \text{ (or 15.42\%)}$$

# Question 3

- (a) A company expects a net operating income of Rs.80,000. It has ₹2,00,000, 8% Debentures. The equity capitalization rate of the company is 10%. Calculate the value of the firm and overall capitalization rate according to the Net Income Approach.
- (b) If the debenture debt is increased to ₹ 6,00,000, what shall be the value of the firm and the overall capitalization rate?

Solution: (a)

Net Income = EBIT - Interest

Net Income = 80,000 - (0.08 x 2,00,000) = 80,000 - 16,000 = 64,000

Value of Equity = Net Income / Cost of Equity ( $k_e$ )

Value of Equity = 64,000 / 0.1 = 6,40,000

Value of the Firm = Value of Equity + Value of Debt

Value of the Firm = 6,40,000 + 2,00,000

Value of the Firm = 8,40,000

Value of the Firm = EBIT / WACC

WACC = EBIT / Market value of the Firm

WACC = 80,000 / 8,40,000

WACC = 0.0952 (or 9.52%)

**Solution: (b)**

$$\text{Total Debt} = 6,00,000$$

$$\text{Net Income} = \text{EBIT} - \text{Interest}$$

$$\text{Net Income} = 80,000 - (0.08 \times 6,00,000) = 80,000 - 48,000 = 32,000$$

$$\text{Value of Equity} = \text{Net Income} / \text{Cost of Equity (ke)}$$

$$\text{Value of Equity} = 32,000 / 0.1 = 3,20,000$$

$$\text{Value of the Firm} = \text{Value of Equity} + \text{Value of Debt}$$

$$\text{Value of the Firm} = 3,20,000 + 6,00,000$$

$$\text{Value of the Firm} = 9,20,000$$

$$\text{Value of the Firm} = \text{EBIT} / \text{WACC}$$

$$\text{WACC} = \text{EBIT} / \text{Market value of the Firm}$$

$$\text{WACC} = 80,000 / 9,20,000$$

$$\text{WACC} = 0.0870 \text{ (or 8.70\%)}$$

|                   | Scenario (a) | Scenario (b) |
|-------------------|--------------|--------------|
| EBIT              | 80,000       | 80,000       |
| Interest          | 16,000       | 48,000       |
| Net Income        | 64,000       | 32,000       |
|                   |              |              |
| Value of Equity   | 6,40,000     | 3,20,000     |
|                   |              |              |
| Value of the Firm | 8,40,000     | 9,20,000     |
|                   |              |              |
| WACC              | 0.0952       | 0.0870       |

# Traditional Approach

Optimal debt-equity mix for a company is the one which has lowest WACC

Assumptions:

- Cost of Debt ( $K_d$ ) and Cost of Equity ( $K_e$ ) are not constant
- $K_d < K_e$

## Step 1

Calculate Operating Profits (EBIT)

## Step 2

Calculate Net Income

## Step 3

Calculate Value of Equity

Value of Equity = Net Income /  $K_e$

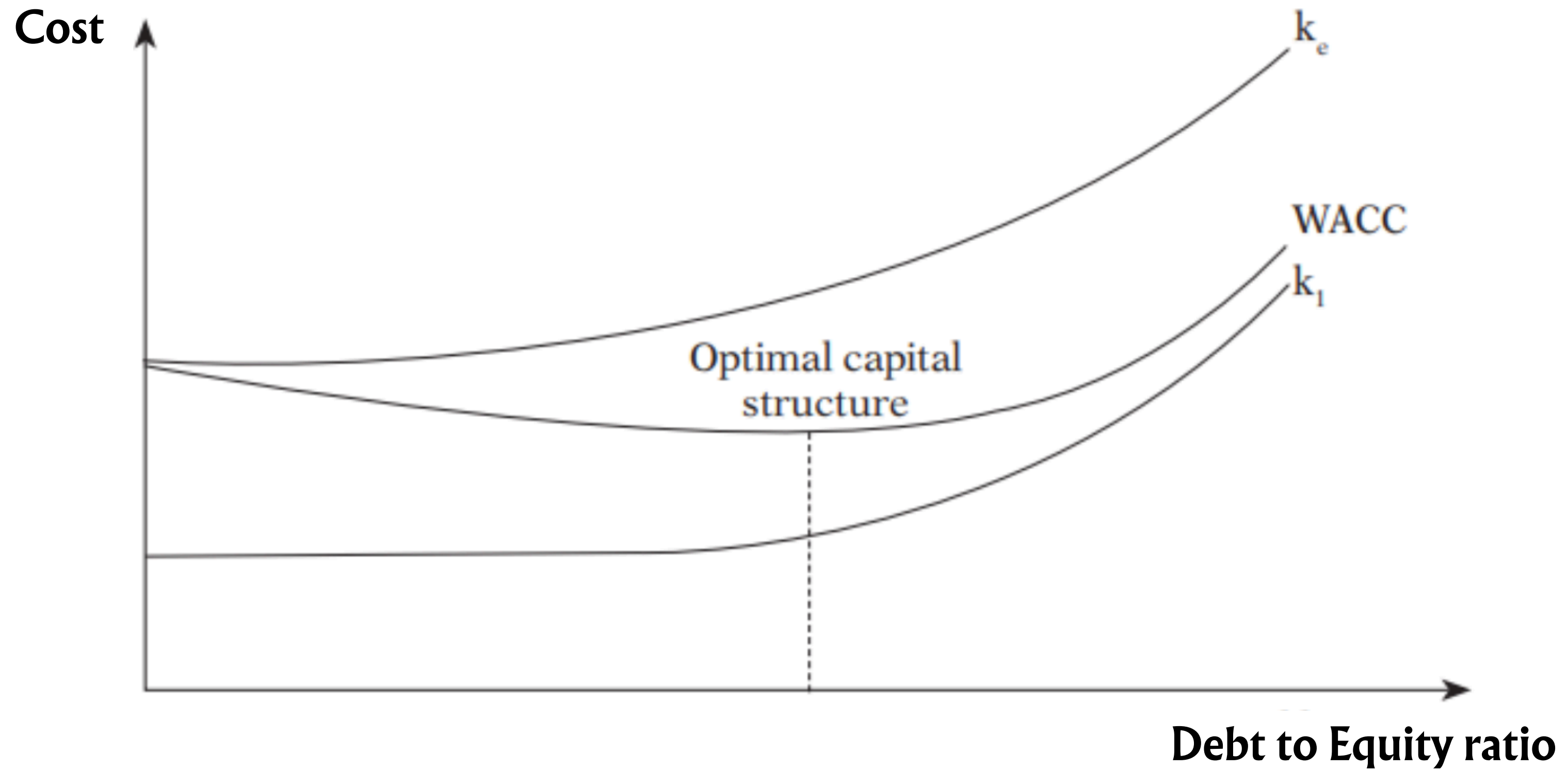
## Step 4

Calculate Value of the Firm

Value of the Firm = Value of Equity + Debt

OR

Value of the Firm = EBIT / WACC



# Traditional Approach

# Question 4

Compute the market value of the firm, value of shares and the average cost of capital from the following information:

Net Operating Income = ₹2,00,000

Equity Capitalization Rate :

- |                                             |     |
|---------------------------------------------|-----|
| (a) If the firm uses no debt -              | 10% |
| (b) If the firm uses ₹4,00,000 debentures - | 11% |
| (c) If the firm uses ₹6,00,000 debenture -  | 13% |

Assume that ₹ 4,00,000 debentures can be raised at 5% rate of interest whereas ₹6,00,000 debentures can be raised at 6% rate of interest.

|                                                         | Scenario (a)     | Scenario (b)     | Scenario (c)     |
|---------------------------------------------------------|------------------|------------------|------------------|
| EBIT                                                    | 2,00,000         | 2,00,000         | 2,00,000         |
| Interest = Debt x Cost of debt                          | 0                | 20,000           | 36,000           |
| <b>Net Income = EBIT - Interest</b>                     | <b>2,00,000</b>  | <b>1,80,000</b>  | <b>1,64,000</b>  |
|                                                         |                  |                  |                  |
| Cost of Equity                                          | 0.10             | 0.11             | 0.13             |
| <b>Value of Equity = Net Income / Cost of Equity</b>    | <b>20,00,000</b> | <b>16,36,364</b> | <b>12,61,538</b> |
| Value of debt                                           | 0                | 4,00,000         | 6,00,000         |
| <b>Total value of the Firm = Value of Equity + Debt</b> | <b>20,00,000</b> | <b>20,36,364</b> | <b>18,61,538</b> |
|                                                         |                  |                  |                  |
| <b>WACC = EBIT / Total value of the Firm</b>            | <b>0.1000</b>    | <b>0.0982</b>    | <b>0.1074</b>    |

# Question 5

A company's current net operating income (EBIT) is ₹8,00,000. The company has ₹20 lakhs of 10% debt outstanding. Its equity capitalization rate is 15%.

The company is considering to increase its debt by raising additional ₹ 10 lakhs and to utilize these funds to retire the amount of equity. However, due to increased financial risk, the cost of entire debt is likely to increase to 12% and the cost of equity it 18%.

You are required to compute the market value of the company using traditional model and also make recommendations regarding the proposal.

|                                                         | Scenario (a)     | Scenario (b)     |
|---------------------------------------------------------|------------------|------------------|
| EBIT                                                    | 8,00,000         | 8,00,000         |
| Interest = Debt x Cost of debt                          | 2,00,000         | 3,60,000         |
| <b>Net Income = EBIT - Interest</b>                     | <b>6,00,000</b>  | <b>4,40,000</b>  |
|                                                         |                  |                  |
| Cost of Equity                                          | 0.15             | 0.18             |
| <b>Value of Equity = Net Income / Cost of Equity</b>    | <b>40,00,000</b> | <b>24,44,444</b> |
| Value of debt                                           | 20,00,000        | 30,00,000        |
| <b>Total value of the Firm = Value of Equity + Debt</b> | <b>60,00,000</b> | <b>54,44,444</b> |
|                                                         |                  |                  |
| <b>WACC = EBIT / Total value of the Firm</b>            | <b>0.1333</b>    | <b>0.1469</b>    |

**Step 1**

**Calculate Operating Profits (EBIT)**

**Step 2**

**Calculate Value of the Firm**

**Value of the Firm =  $EBIT / WACC$**

**Step 3**

**Calculate Value of Equity**

**Value of Equity = Value of the Firm - Debt**

**Step 4**

**Calculate Cost of Equity**

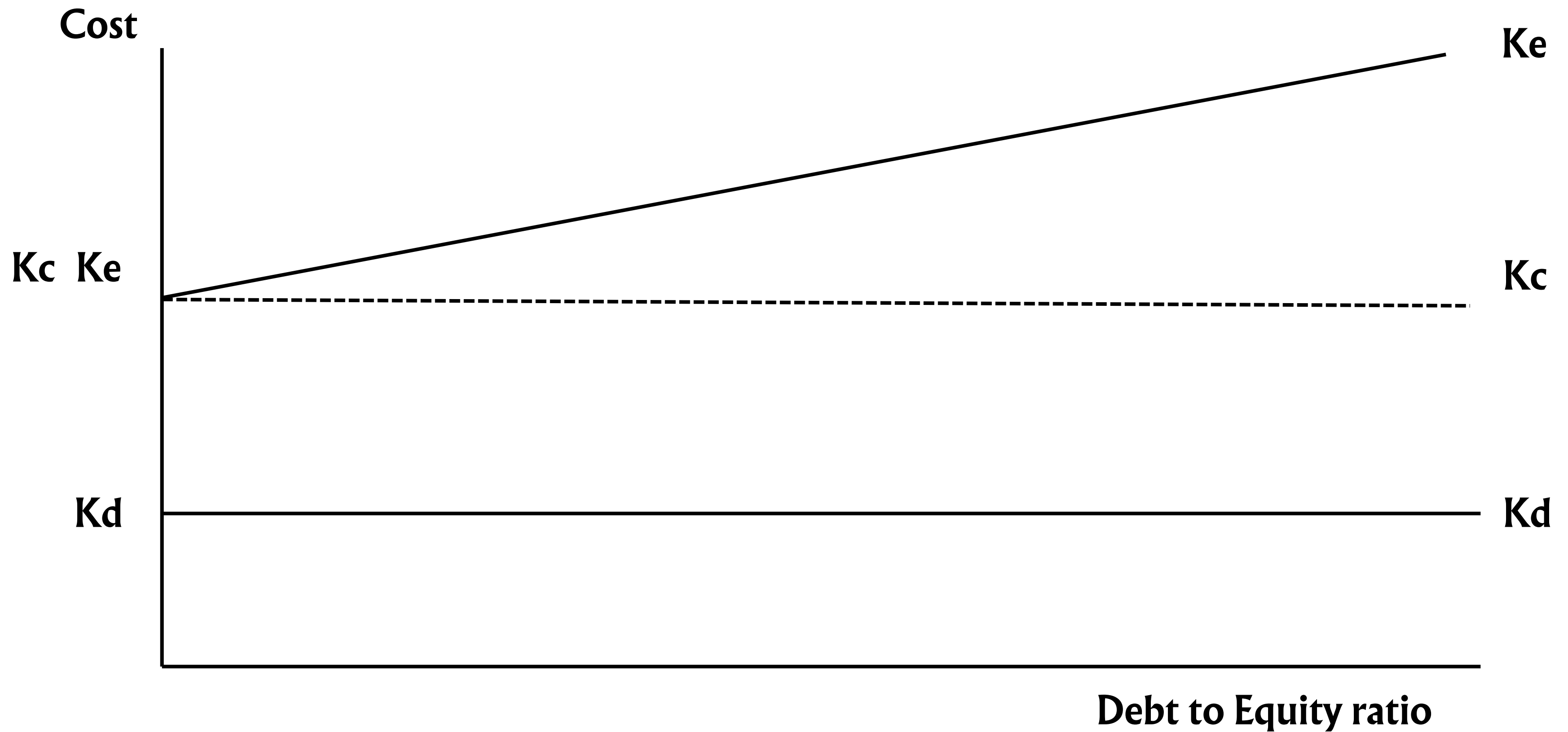
**Cost of Equity =  $Net\ Income / Value\ of\ Equity$**

# Net Operating Income Approach

Capital Structure composition has no effect on the value of the firm since WACC is constant

**Assumptions:**

- Cost of debt < Cost of equity
- Cost of debt is constant
- WACC is constant
- Cost of equity increase with increase in debt



# Net Operating Income Approach

# Question 6

**X Ltd has earned a profit of Rs.500,000 before interest and tax. The company's capital structure includes 20,000, 14% debentures of Rs 100 each. The overall capitalization rate of the firm is 16%. Compute value of the firm and equity capitalization rate using NOI approach.**

Solution:

$$\text{Value of the Firm} = \text{EBIT} / \text{WACC}$$

$$\text{Value of the Firm} = 500,000 / 0.16$$

$$\text{Value of the Firm} = 31,25,000$$

$$\text{Value of Debt} = 20,000 \times 100 = 20,00,000$$

$$\text{Value of Equity} = \text{Value of the Firm} - \text{Value of Debt}$$

$$\text{Value of Equity} = 31,25,000 - 20,00,000$$

$$\text{Value of Equity} = 11,25,000$$

$$\text{Net Income} = \text{EBIT} - \text{Interest}$$

$$\text{Net Income} = 5,00,000 - (0.14 \times 20,00,000) = 5,00,000 - 2,80,000 = 2,20,000$$

$$\text{Cost of Equity (ke)} = \text{Net Income} / \text{Value of Equity}$$

$$\text{Cost of Equity (ke)} = 2,20,000 / 11,25,000$$

$$\text{Cost of Equity (ke)} = 0.1956 \text{ (or 19.56\%)}$$

# Question 7

Summer Ltd expects annual operating income of Rs.2,00,000. It has Rs.5,00,000 outstanding debt, cost of debt is 10%. If the overall capitalization rate is 12.5%, what would be the total value of the firm and the equity capitalization rate according to the NET Operating Income approach.

What will be the effect of the following on the total value of the firm and equity capitalization rate, if:

1. The firm increases the amount of debt from Rs.5,00,000 to Rs.7,50,000 and uses the proceeds of the debt to repurchase equity shares.
2. The firm redeems debt of Rs.2,50,000 by issuing fresh equity shares of the same amount.

|                                                      | Current          | Scenario 1       | Scenario 2       |
|------------------------------------------------------|------------------|------------------|------------------|
| EBIT                                                 | 2,00,000         | 2,00,000         | 2,00,000         |
| WACC                                                 | 0.125            | 0.125            | 0.125            |
| <b>Value of the Firm = EBIT / WACC</b>               | <b>16,00,000</b> | <b>16,00,000</b> | <b>16,00,000</b> |
| Value of Debt                                        | 5,00,000         | 7,50,000         | 2,50,000         |
| <b>Value of Equity = Value of the Firm - Debt</b>    | <b>11,00,000</b> | <b>8,50,000</b>  | <b>13,50,000</b> |
|                                                      |                  |                  |                  |
| EBIT                                                 | 2,00,000         | 2,00,000         | 2,00,000         |
| Interest                                             | 50,000           | 75,000           | 25,000           |
| <b>Net Income = EBIT - Interest</b>                  | <b>1,50,000</b>  | <b>1,25,000</b>  | <b>1,75,000</b>  |
|                                                      |                  |                  |                  |
| <b>Cost of Equity = Net Income / Value of Equity</b> | <b>0.1364</b>    | <b>0.1471</b>    | <b>0.1296</b>    |

### Step 1

Calculate Operating Profits (EBIT)

### Step 2

Calculate Value of the Firm

Value of the unlevered firm = EBIT / WACC

Value of levered firm = Value of unlevered firm

### Step 3

Calculate Value of Equity

Value of Equity = Value of the Firm - Debt

### Step 4

Calculate Leverage Cost of Equity

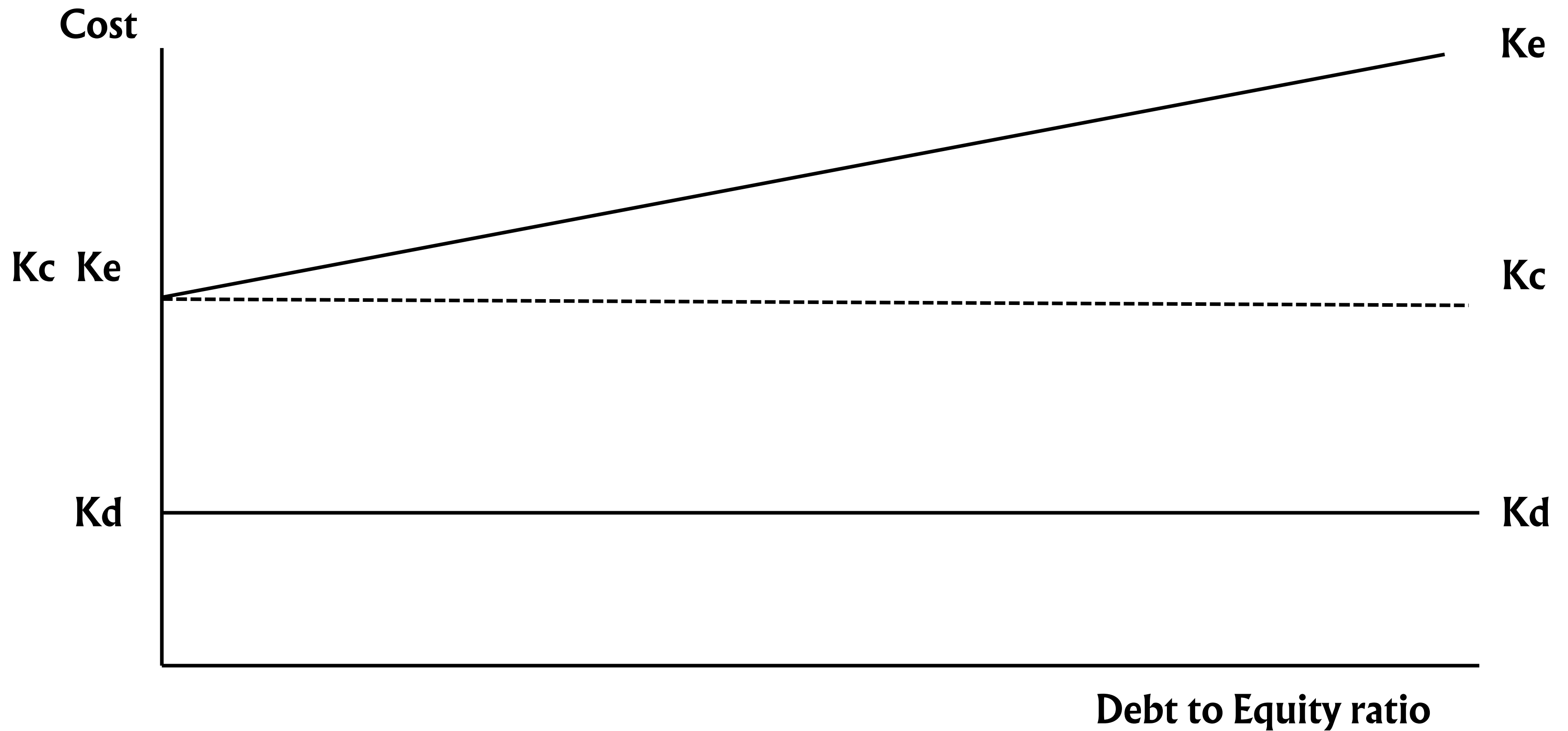
$$K_{eL} = K_{eu} + (K_{eu} - K_d) \times \frac{D}{E}$$

# MM Approach (Without tax)

Overall cost of capital and value of the firm are independent of its Capital Structure.

Assumptions:

- Capital markets are perfect
- All investors are rational
- Investors can borrow and lend at the risk-free rate
- Non-existence of corporate taxes



# MM Approach (Without tax)

# Question 8

Mid Air enterprises is an all equity company. The expected EBIT of the company is Rs.24 lakhs.

If the company begins to use debt, it can borrow at the rate of 8%. This rate is independent of the amount of debt. Any money raised by selling debt would be used to retire common stock.

The equity capitalization rate would be 12% when no debt is used. Assuming a debt of 1 crore, using MM model without corporate taxes:

- a) Determine the firm's total market value
- b) Determine the firm's value of equity
- c) Determine the firm's leverage cost of equity.

### Solution:

a) Value of the Firm = EBIT / WACC

In case of no debt, WACC =  $K_e$ . Thus,

Value of the Firm = 24,00,000 / 0.12

**Value of the Firm = 2,00,00,000**

b) Value of equity in a leveraged firm is,

Value of Equity = Value of the Firm - Value of Debt

Value of Equity = 2,00,00,000 - 1,00,00,000

**Value of Equity = 1,00,00,000**

c) Firm's leverage cost of equity is,

$$K_{eL} = K_{eu} + (K_{eu} - K_d) \times \frac{D}{E}$$

$K_{eL} = 0.12 + (1/1) \times (0.12 - 0.08)$

$K_{eL} = 0.12 + 0.04$

**$K_{eL} = 0.16$  (or 16%)**

|                                     | Unlevered          | Levered            |
|-------------------------------------|--------------------|--------------------|
| EBIT                                | 24,00,000          | 24,00,000          |
| Debt                                | 0                  | 1,00,00,000        |
| Cost of debt                        | 0                  | 0.08               |
| Cost of equity                      | 0.12               |                    |
|                                     |                    |                    |
| EBIT                                | 24,00,000          | 24,00,000          |
| Interest = Cost of debt x debt      | -                  | 8,00,000           |
| <b>Net Income = EBIT - Interest</b> | <b>24,00,000</b>   | <b>16,00,000</b>   |
|                                     |                    |                    |
| Value of Debt                       | -                  | 1,00,00,000        |
| Value of Equity                     | 2,00,00,000        | 1,00,00,000        |
| <b>Value of the Firm</b>            | <b>2,00,00,000</b> | <b>2,00,00,000</b> |
|                                     |                    |                    |
| <b>Cost of equity</b>               | <b>12.00%</b>      | <b>16.00%</b>      |
| <b>WACC</b>                         | <b>12.00%</b>      | <b>12.00%</b>      |

**Step 1**

Calculate Operating Profits (EBIT)

**Step 2**

Calculate Value of the Firm

Value of the unlevered Firm =  $EBIT \times (1 - T) / WACC$

Value of levered firm = Value of unlevered firm + Tax benefits

(Tax benefits = Value of Debt x Tax rate)

**Step 3**

Calculate Value of Equity

Value of Equity = Value of the Firm - Debt

**Step 4**

Calculate Leverage Cost of Equity

$$K_{eL} = K_{eu} + (K_{eu} - K_d) \times \frac{D}{E} \times (1 - T)$$

# MM Approach (With tax)

Due of the tax advantages of issuing debt finance, firms should increase their gearing as much as possible.

Assumptions:

- Capital markets are perfect
- All investors are rational
- Firms are grouped into 'Equivalent Risk Classes' on the basis of business risk

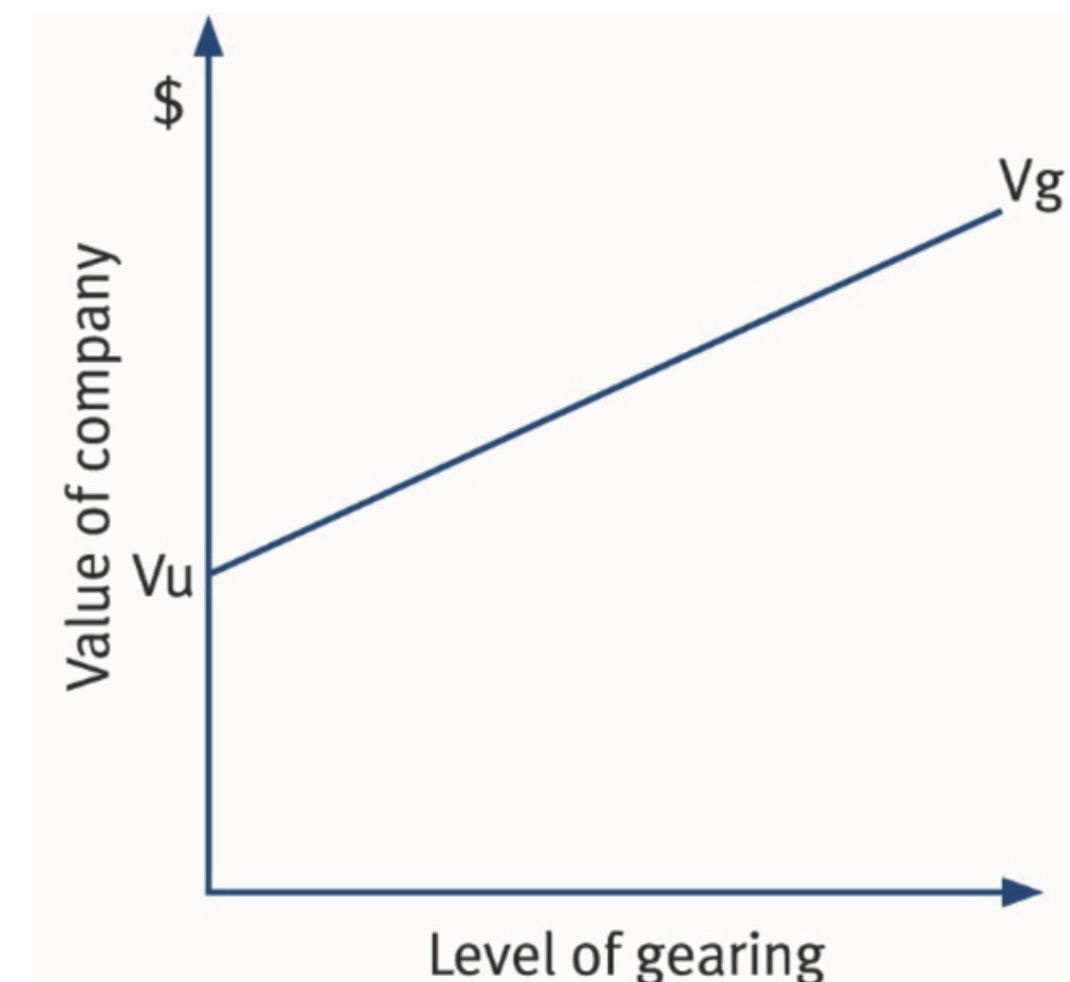
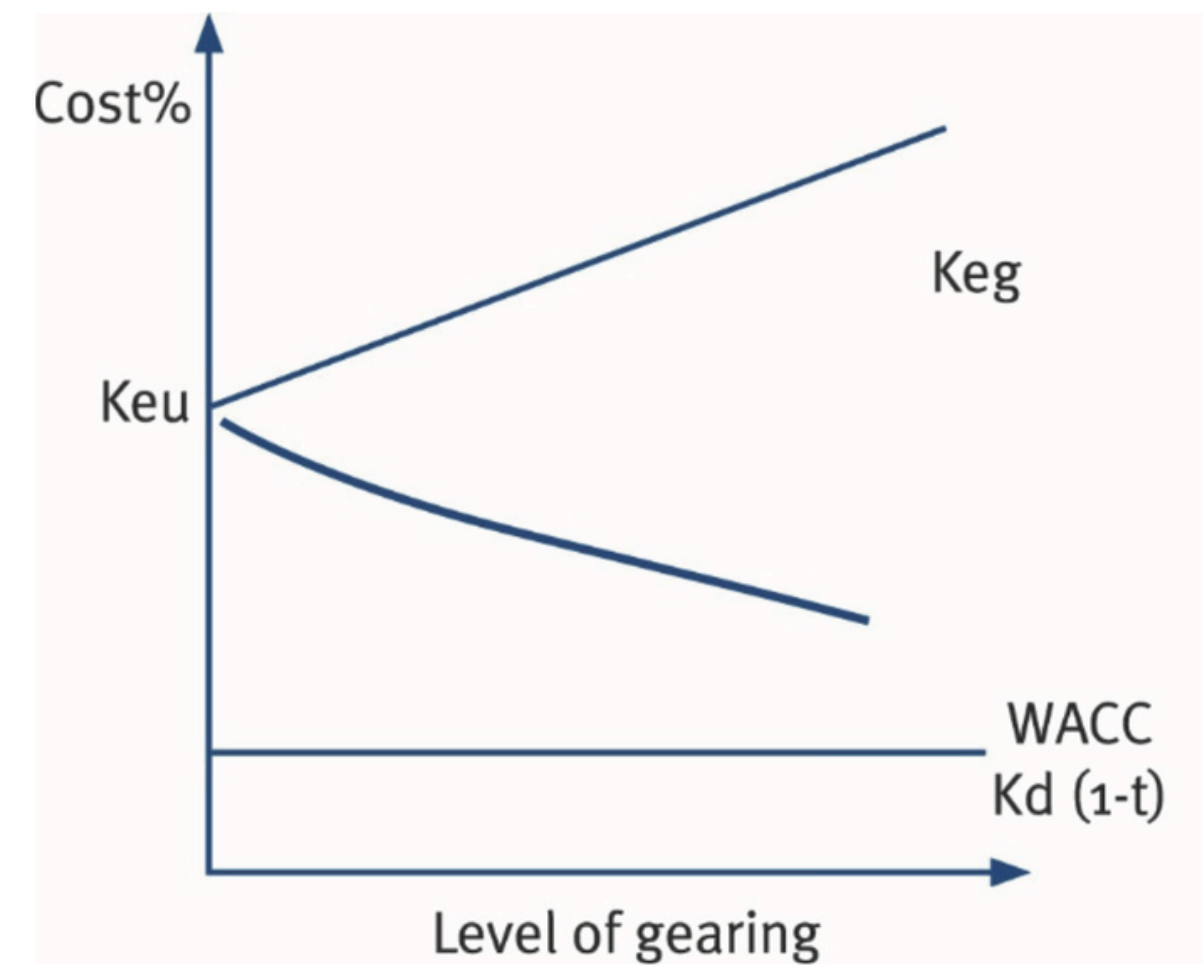
As investors are rational, the  $K_e$  is directly linked to the increase in gearing

However:

- Debt interest is tax deductible  $\rightarrow K_d$  is lower than in MM (no tax)
- Lower debt costs imply less volatility in returns for the same level of gearing, giving smaller increases in  $K_e$
- The increase in  $K_e$  does not offset the benefit of the cheaper debt finance  $\rightarrow$  WACC falls as gearing is increased

Conclusion:

- Gearing up reduces the WACC, and the optimal capital structure is 99.9% gearing.



# Practical Arguments Against MM

- **Bankruptcy risk**
- **Agency costs: restrictive conditions**
- **Tax exhaustion**
- **Borrowing/debt capacity**
- **Difference of risk tolerance levels between shareholders and directors**
- **Increases in the cost of borrowing as gearing increases**

# Question 9

X and Y are two firms identical in all aspects except that X does not use any debt in its financing, while Y has Rs.1,00,000 5% debentures in its financing. Both the firms have earnings before interest and tax of Rs.25,000 and the equity capitalization rate is 10%. Assuming the corporate tax to be 50%, calculate the value of the firm using M&M approach.

## Solution:

For Firm X,

$$\text{EBIT} = 25,000$$

$$\text{Net Income} = \text{EBIT} \times (1 - T) = 25,000 \times (1 - 0.5) = 12,500$$

$$\text{Value of Equity} = \text{Net Income} / K_e = 12,500 / 0.1 = 125,000$$

$$\text{Value of the Firm X} = \text{Debt} + \text{Equity} = 0 + 125,000 = 125,000$$

$$\text{Value of the firm Y} = 125,000 + (1,00,000 \times 0.05 \times 0.5) = 125,000 + 25,000 = \underline{\underline{150,000}}$$

# Question 10

ABC Ltd. has a net operating income of Rs.3,00,000. If equity capitalization rate is 18%. Compute the value of the firm using MM (no leverage, no tax scenario). Recompile the value of the firm under the modern approach if tax rate is 40%.

## Solution:

MM (no leverage, no tax)

EBIT = 3,00,000

Ke = 18%

Value of the Firm = EBIT / WACC

In case of no debt, WACC = Ke. Thus,

Value of the Firm = 3,00,000 / 0.18

Value of the Firm = 16,66,667

MM (no leverage, with tax)

EBIT = 3,00,000

Ke = 18%

Tax rate = 40%

Value of the Firm = EBIT x (1 - T) / WACC

In case of no debt, WACC = Ke. Thus,

Value of the Firm = 3,00,000 x (1 - 0.4) / 0.18

Value of the Firm = 1,80,000 / 0.18

Value of the Firm = 10,00,000

# Question 11

XYZ Ltd has a net operating income of Rs.700,000. Its equity capitalization rate is 12% with a taxation rate of 30%. What would be its value? If XYZ goes in for a debt of Rs.400,000 with an interest rate of 10%, will the value of the firm change.

## Solution:

MM (no leverage, with tax)

EBIT = 7,00,000

$K_e = 12\%$

Tax rate = 40%

Value of the Firm =  $EBIT \times (1 - T) / WACC$

In case of no debt,  $WACC = K_e$ . Thus,

Value of the Firm =  $7,00,000 \times (1 - 0.3) / 0.12$

Value of the Firm =  $4,90,000 / 0.12$

Value of the Firm = 40,83,333

MM (with leverage, with tax)

Debt = 4,00,000

Tax rate = 30%

Tax benefits = Debt x Tax rate

Tax benefits =  $4,00,000 \times 0.3 = 1,20,000$

Value of levered firm = Value of unlevered firm + Tax benefits

Value of the levered Firm =  $40,83,333 + 1,20,000$

Value of the levered Firm = 42,03,333

# Question 12

ABC Ltd has a cost of debt and overall cost of capital at 9% and 12% respectively. The ratio of market value of debt to equity is 0.8. What will be the expectation of the Equity shareholders from this firm if shareholders of unlevered firm expect 14% as rate of return.

## Solution:

$$K_{eu} = 14\%$$

$$K_d = 9\%$$

$$WACC = 12\%$$

$$\text{Debt} / \text{Equity} = 0.8$$

$$K_{eL} = K_{eu} + (K_{eu} - K_d) \times \frac{D}{E}$$

Firm's leverage cost of equity is,

$$K_{eL} = 0.14 + (0.14 - 0.09) \times 0.8$$

$$K_{eL} = 0.14 + 0.05 \times 0.8$$

$$K_{eL} = 0.14 + 0.04$$

$$K_{eL} = 0.18 \text{ (or 18\%)}$$

# Question 13

XYZ identified a project with 100% equity and a cost of 21%. ABC Ltd., another company, identical to XYZ Ltd has a capital structure of 2:1 in debt equity ratio.  $K_d$  of this company is 14%. Compute  $K_e$  of ABC Ltd and prove that WACC is constant.

## Solution:

$$K_{eu} = 21\%$$

$$K_d = 14\%$$

$$WACC = 21\%$$

$$\text{Debt / Equity} = 2$$

Firm's leverage cost of equity is,

$$K_{eL} = K_{eu} + (K_{eu} - K_d) \times \frac{D}{E}$$

$$K_{eL} = 0.21 + (0.21 - 0.14) \times 2$$

$$K_{eL} = 0.35 \text{ (or 35\%)}$$

$$WACC = \left( \frac{V_e}{V_e + V_d} \right) k_e + \left( \frac{V_d}{V_e + V_d} \right) k_d (1 - T)$$

$$WACC = 1/(1+2) \times 0.35 + 2/(1+2) \times 0.14$$

$$WACC = 1/3 \times 0.35 + 2/3 \times 0.14$$

$$WACC = 0.1167 + 0.0933$$

$$WACC = 0.21 \text{ (or 21\%)}$$

